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Technical Document

EMV® Transaction Processing Quick Reference Guide

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Rapid Connect

Table of Contents

Chapter 1 Introduction 4
1.1 Overview 4
Chapter 2 Supported Payment and Transaction Types 5
Chapter 3 EMV® Processing Requirements 6
3.1 Contactless Technology for EMV® 6
Chapter 4 EMV® Transaction Process Flow 7
4.1 EMV® Contact Transaction 7
4.2 EMV® Contactless Transaction 7
Chapter 5 EMV® Receipt Requirements 8
Chapter 6 EMV® Fallback 9
6.1 EMV® Fallback Scenarios 9
6.2 Risks in EMV® Fallback 9

Revision History

Section	Change Log	Version	Date
2	Added Visa Fleet, Wright Express, Voyager, Fleet One, and FleetCor to list of supported payment types and added corresponding footnote.	12.04.5	October 26, 2022
5	Updated the description to the following: "For more information regarding Receipt Requirements and EMV in general, please refer to the Fiserv EMV Implementation Guide stored in the Fiserv Merchant Services® Specification and EMV Implementation Guide portal. To access the portal, please go to: https://specs.firstdata.com/specs/kb/login/index_enter/. While the Rapid Connect Specification is not housed at the aforementioned portal, you will find links to other specifications and platforms currently supported by Fiserv. You will need to be registered to access the specification documents. Please be sure to Subscribe for any document listed in order to receive notifications through email when such document is updated. The Fiserv EMV Implementation Guide will provide insight on the Fiserv EMV process. To access the Fiserv Generic EMV Implementation Guide once registered: • Under Specifications Description, select the First Data Merchant Services® Folder • Select EMV Folder from Sub Categories • Select Implementation Guide • Select Fiserv Generic EMV Implementation Guide and Download • Accept FDMS Specifications License Agreement"	9.02.18	January 7, 2020

Chapter 1 Introduction

1.1 Overview

EMV is a global standard of credit card payment, based on chip card technology. EMV cards, also known as smart cards, have an embedded microprocessor chip. EMV cards support both contact and contactless payments.

For more information on EMV, please refer to http://www.emvco.com.

Chapter 2 Supported Payment and Transaction Types

The following table lists Payment Types supported by the Rapid Connect UMF:

Payment Type	Supported Card Types	Supported Transaction Types	Timeout Reversal and Void
Credit	Visa® Mastercard® American Express® Discover® Network Card Types	Authorization Completion Sale BatchSettleDetail Refund Verification Balance Inquiry[1]	Timeout Reversal Supported Void/Full Reversal Supported[*1]
Debit		Authorization[3] Completion[4] Sale BatchSettleDetail Refund Balance Inquiry	Timeout Reversal Supported Void/Full Reversal Supported[*1]
Private Label		Authorization Completion Sale BatchSettleDetail Refund Verification	Timeout Reversal Supported Void/Full Reversal Supported[*1]
Fleet	Visa Fleet [*2] Wright Express Voyager Fleet One FleetCor	Authorization Completion Sale Refund	Timeout Reversal Supported Void/Full Reversal Supported[*1]

[1] For Credit, Balance Inquiry transactions are only supported when the Card Type is Mastercard

[2] Applicable to Visa Fleet 2.0

[3] Only supported for merchants sending the Settlement Indicator value of 1 or 2 in the transaction, Petroleum merchants, and some Host Capture merchants. For more information please contact your account representative

[4] Only supported when the Settlement Indicator field is not present in a Credit Authorization transaction that included the PINLess POS Debit Flag field, the EMVData field with Tag 9F06 and/or 84, and only when the PINLess POS Debit Flag field. value of "D" (Processed as Debit) was returned in the response message

Chapter 3 EMV® Processing Requirements

3.1 Contactless Technology for EMV®

Rapid Connect supports the following Card Types and Transaction Types for EMV contactless transactions:

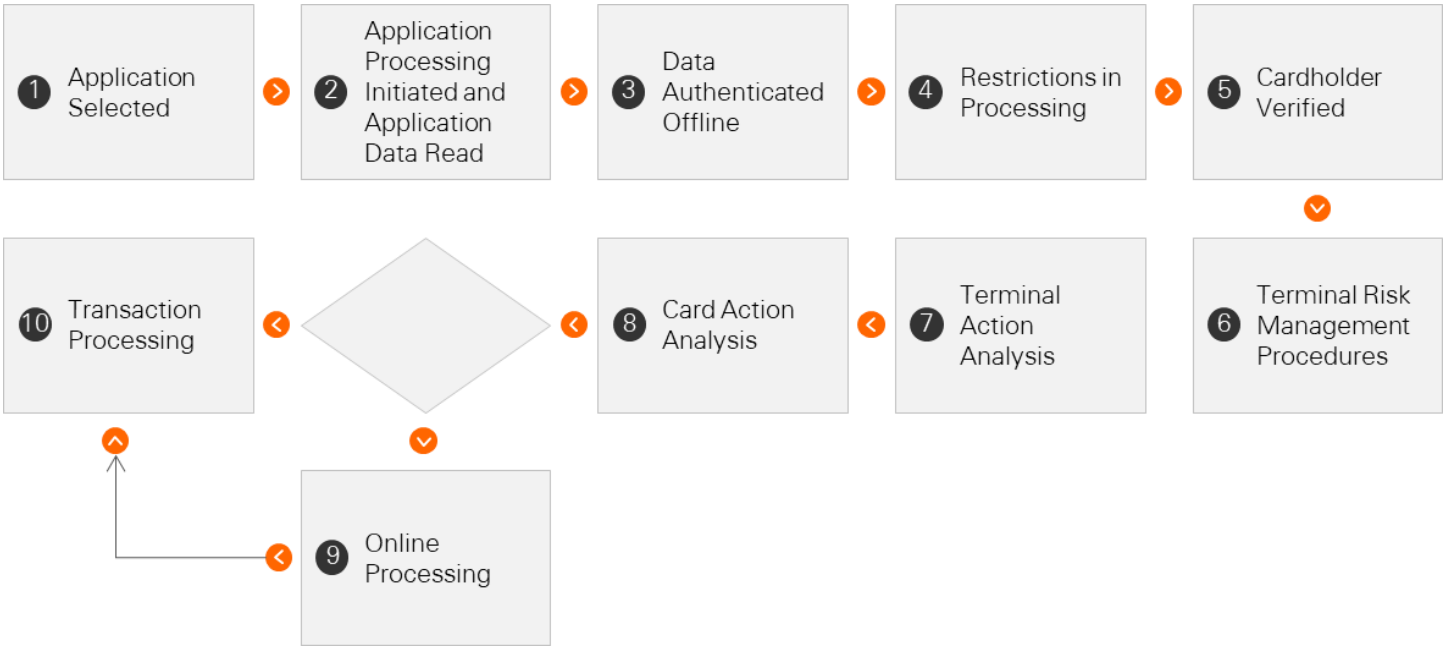
Payment Type	Supported Card Types	Supported Transaction Types
Credit	Visa Mastercard American Express Discover Network (including JCB and Diners)	Authorization Completion Sale BatchSettleDetail Refund
Debit		Authorization[3] Completion[4] Sale BatchSettleDetail Refund

[3] Only supported for merchants sending the Settlement Indicator value of 1 or 2 in the transaction, Petroleum merchants, and some Host Capture merchants. For more information please contact your account representative

4 Only supported when the Settlement Indicator field is not present in a Credit Authorization transaction that included the PINLess POS Debit Flag field, the EMVData field with Tag 9F06 and/or 84, and only when the PINLess POS Debit Flag field. value of "D" (Processed as Debit) was returned in the response message

Chapter 4 EMV® Transaction Process Flow

4.1 EMV® Contact Transaction



1. EMV application selected
2. Application is initiated and necessary data is read from chip by the terminal
3. Data authenticated offline
4. Chip's permission to execute transaction is checked
5. Cardholder verification
6. Terminal conducts checks like floor limit, for online processing requirement
7. Based on results from above verification steps, terminal application requests: a decline offline/approve offline/go online result
8. Chip's response with decline offline/approve offline/go online, based on issuer defined limits
9. For go online request, terminal requests Issuer for authorization and online card authentication
10. Transaction processed and completed

4.2 EMV® Contactless Transaction

In contactless EMV transactions some of the transaction steps may be executed after the chip has left the proximity of the card reader (online authorization).

Chapter 5 EMV® Receipt Requirements

For more information regarding Receipt Requirements and EMV in general, please refer to the Fiserv EMV Implementation Guide stored in the Fiserv Merchant Services® Specification and EMV Implementation Guide portal.

To access the portal, please go to: https://specs.firstdata.com/specs/kb/login/index_enter/.

While the Rapid Connect Specification is not housed at the aforementioned portal, you will find links to other specifications and platforms currently supported by Fiserv. You will need to be registered to access the specification documents. Please be sure to SUBSCRIBE for any document listed in order to receive notifications through email when such document is updated. The Fiserv EMV Implementation Guide will provide insight on the Fiserv EMV process.

To access the Fiserv Generic EMV Implementation Guide once registered:

- Under Specifications Description, select the First Data Merchant Services® Folder
- Select EMV Folder from Sub Categories
- Select Implementation Guide
- Select Fiserv Generic EMV Implementation Guide and Download
- Accept FDMS Specifications License Agreement

Chapter 6 EMV® Fallback

Most EMV cards contain a magnetic stripe, to support fallback if the EMV enabled chip is unreadable. Fallback is a non EMV chip transaction that is performed with EMV chip cards at the EMV capable devices. If an EMV chip card is accepted at a terminal that is not EMV capable, it is considered as a magnetic stripe transaction instead of a fallback transaction. It would only be considered a fallback transaction if the terminal was EMV capable but could not read the EMV chip data on the card.

6.1 EMV® Fallback Scenarios

Following are the scenarios which lead to EMV fallback transactions:

- **Damaged chip card/Dirty card reader**: Even if both the terminal and card are fully EMV compliant, it may not be possible for the EMV card reader to communicate with the chip, for example, if the chip on the card has become accidentally damaged or if the contacts on the card or reader are dirty
- **Issues in interoperability**: Interoperability issues may prevent the EMV transaction from being completed

6.2 Risks in EMV® Fallback

Following are the risks involved in EMV fallback transactions:

- Chances of fraud are high in fallback transactions. Fraudsters, often, deliberately disable the chip to try and circumvent the ability to verify the cardholder using their PIN
- Unattended devices should not support fallback transactions as a card may be deliberately damaged and there are no merchants available to check the damage on the card

Connect With Us

For more information about EMV® Transaction Processing Quick Reference Guide:

800-872-7882

getsolutions@fiserv.com

fiserv.com

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Fiserv, Inc.
255 Fiserv Drive
Brookfield, WI 53045
fiserv.com

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